

```
!pip install FlagEmbedding
```

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```

```
from FlagEmbedding import BGEM3FlagModel

model = BGEM3FlagModel('BAAI/bge-m3',
                        use_fp16=True) # Setting use_fp16 to True speeds up computation with a slight performance degradation

sentences_1 = ["What is BGE M3?", "Defination of BM25"]
sentences_2 = ["BGE M3 is an embedding model supporting dense retrieval, lexical matching and multi-vector interaction.",
               "BM25 is a bag-of-words retrieval function that ranks a set of documents based on the query terms appearing in e

embeddings_1 = model.encode(sentences_1,
                             batch_size=12,
                             max_length=8192, # If you don't need such a long length, you can set a smaller value to speed up th
                             )['dense_vecs']
embeddings_2 = model.encode(sentences_2)['dense_vecs']
similarity = embeddings_1 @ embeddings_2.T
print(similarity)
# [[0.6265, 0.3477], [0.3499, 0.678 ]]
```

```

Download complete: :      0.00B

Reconstruction complete:      0.00B /  0.00B

Fetching 30 files: 100%                                30/30 [00:00<00:00, 942.12it/s]

Loading weights: 100%                                391/391 [00:08<00:00, 55.07it/s]
[[0.6259035  0.3474958 ]
 [0.34986812 0.6782463  ]]

```

```

from FlagEmbedding import BGEM3FlagModel

model = BGEM3FlagModel('BAAI/bge-m3',
                        use_fp16=True)

```

```

Download complete: :      0.00B

Reconstruction complete:      0.00B /  0.00B

Fetching 30 files: 100%                                30/30 [00:00<00:00, 361.37it/s]

Loading weights: 100%                                391/391 [00:04<00:00, 129.18it/s]

```

```

sentences_1 = ["I LOVE GENAI"]
embeddings_1 = model.encode(sentences_1,
                             batch_size=12,
                             max_length=8192, # If you don't need such a long length, you can set a smaller value to speed up the
                             )['dense_vecs']

```

```
embeddings_1
```

```
array([[ -0.04204996,  0.0040168 , -0.01458733, ..., -0.02487984,
        -0.04212679,  0.03624649]], dtype=float32)
```

```

sentences_2 = ["I LOVE GENAI"]
embeddings_2 = model.encode(sentences_2,
                             batch_size=12,
                             max_length=8192, # If you don't need such a long length, you can set a smaller value to speed up the
                             )['dense_vecs']

```

```
embeddings_2
```

```
array([[ -0.04204996,  0.0040168 , -0.01458733, ..., -0.02487984,
        -0.04212679,  0.03624649]], dtype=float32)
```

```

import numpy as np

#cosine similarity

similarity = np.dot(embeddings_1[0], embeddings_2[0]) / (np.linalg.norm(embeddings_1[0]) * np.linalg.norm(embeddings_2[0]))

```

```
similarity
```

```
np.float32(1.0)
```

```
#RECOMMENDATION SYSTEM MOVIES : BASED GENRE
```

```

movies = [
    {
        "title": "Inception",
        "genre": "Sci-Fi, Thriller",
        "description": "A skilled thief enters people's dreams to steal secrets and attempts to plant an idea in someone's mind
    },
    {
        "title": "Interstellar",
        "genre": "Sci-Fi, Adventure, Drama",
        "description": "Astronauts travel through a wormhole in space to find a new home for humanity."
    },
    {
        "title": "The Dark Knight",
        "genre": "Action, Crime, Drama",

```

```

        "description": "Batman faces a dangerous criminal mastermind who creates chaos and terror across Gotham City."
    },
    {
        "title": "Avengers: Endgame",
        "genre": "Action, Adventure, Sci-Fi",
        "description": "Superheroes reunite to reverse the devastating consequences caused by a powerful villain."
    },
    {
        "title": "Titanic",
        "genre": "Romance, Drama",
        "description": "A young couple from different social backgrounds fall in love during the tragic voyage of the Titanic."
    },
    {
        "title": "The Notebook",
        "genre": "Romance, Drama",
        "description": "Two young people fall deeply in love but face family pressure and separation before finding their way t
    },
    {
        "title": "The Conjuring",
        "genre": "Horror, Mystery, Thriller",
        "description": "Paranormal investigators help a family experiencing terrifying supernatural events in their new home."
    },
    {
        "title": "It",
        "genre": "Horror, Thriller",
        "description": "A group of children confronts a terrifying supernatural creature that appears in the form of a clown."
    },
    {
        "title": "The Hangover",
        "genre": "Comedy",
        "description": "Friends travel to Las Vegas for a bachelor party and wake up with no memory of what happened the previc
    },
    {
        "title": "3 Idiots",
        "genre": "Comedy, Drama",
        "description": "Three engineering students experience friendship, academic pressure, love, and the struggle to follow t
    },
    {
        "title": "Dangal",
        "genre": "Sports, Drama",
        "description": "A former wrestler trains his daughters to become successful wrestlers despite social and family challen
    },
    {
        "title": "The Pursuit of Happyness",
        "genre": "Drama, Biography",
        "description": "A struggling father fights poverty while trying to build a better future for himself and his young son.
    },
    {
        "title": "The Matrix",
        "genre": "Sci-Fi, Action",
        "description": "A computer programmer discovers that reality is a simulated world controlled by intelligent machines."
    },
    {
        "title": "Jurassic Park",
        "genre": "Sci-Fi, Adventure, Thriller",
        "description": "Scientists create a theme park filled with genetically engineered dinosaurs, but the creatures escape a
    },
    {
        "title": "Forrest Gump",
        "genre": "Drama, Romance",
        "description": "A kind-hearted man experiences important historical events while navigating friendship, love, and persc
    }
]

```

```

texts = [
    f"{movie['title']}. Genre: {movie['genre']}. {movie['description']}"
    for movie in movies
]

for text in texts:
    print(text)

```

Inception. Genre: Sci-Fi, Thriller. A skilled thief enters people's dreams to steal secrets and attempts to plant an idea in someone's mind.

Interstellar. Genre: Sci-Fi, Adventure, Drama. Astronauts travel through a wormhole in space to find a new home for humanity.

The Dark Knight. Genre: Action, Crime, Drama. Batman faces a dangerous criminal mastermind who creates chaos and terror across Gotham City.

Avengers: Endgame. Genre: Action, Adventure, Sci-Fi. Superheroes reunite to reverse the devastating consequences caused by the Snap.

Titanic. Genre: Romance, Drama. A young couple from different social backgrounds fall in love during the tragic voyage of the RMS Titanic.

The Notebook. Genre: Romance, Drama. Two young people fall deeply in love but face family pressure and separation before finding each other again.

The Conjuring. Genre: Horror, Mystery, Thriller. Paranormal investigators help a family experiencing terrifying supernatural events.

It. Genre: Horror, Thriller. A group of children confronts a terrifying supernatural creature that appears in the form of a clown.

The Hangover. Genre: Comedy. Friends travel to Las Vegas for a bachelor party and wake up with no memory of what happened the previous night.

3 Idiots. Genre: Comedy, Drama. Three engineering students experience friendship, academic pressure, love, and the struggle to follow their dreams.

Dangal. Genre: Sports, Drama. A former wrestler trains his daughters to become successful wrestlers despite social and family challenges.

The Pursuit of Happyness. Genre: Drama, Biography. A struggling father fights poverty while trying to build a better future for his son.

The Matrix. Genre: Sci-Fi, Action. A computer programmer discovers that reality is a simulated world controlled by intelligent machines.

Jurassic Park. Genre: Sci-Fi, Adventure, Thriller. Scientists create a theme park filled with genetically engineered dinosaurs.

Forrest Gump. Genre: Drama, Romance. A kind-hearted man experiences important historical events while navigating friendship and love.

```
embedding=[]
for text in texts:

    embeddings= model.encode(text,
                              batch_size=12,
                              max_length=8192, # If you don't need such a long length, you can set a smaller value to speed up the encoding
                              )['dense_vecs']
    embedding.append(embeddings)
```

```
len(embedding)
```

```
15
```

```
embedding
```

```
[array([-0.00202693,  0.0201381 , -0.01109634, ..., -0.05026658,
        -0.02285776, -0.01060469], dtype=float32),
 array([-0.02848639, -0.01454587, -0.01094803, ..., -0.02769613,
        -0.01036233, -0.01382132], dtype=float32),
 array([ 0.02084275, -0.00561649, -0.02002954, ..., -0.00626637,
        0.00443491, -0.00091101], dtype=float32),
 array([ 1.7036576e-02,  6.5596098e-05, -2.2263847e-02, ...,
        9.5007792e-03, -1.9547462e-02, -4.3077352e-03], dtype=float32),
 array([-0.01280117,  0.00654286, -0.02925876, ..., -0.00090848,
        -0.00543806, -0.03208977], dtype=float32),
 array([ 0.0002628 ,  0.02121896, -0.02594645, ..., -0.02374038,
        -0.01331762, -0.02248945], dtype=float32),
 array([-0.0235125 ,  0.00596215, -0.05109383, ..., -0.03574776,
        -0.01066542,  0.00150537], dtype=float32),
 array([ 0.00691109,  0.03665004, -0.01400833, ..., -0.02990024,
        -0.0122845 , -0.00668951], dtype=float32),
 array([ 0.01174693,  0.01989347, -0.01785168, ...,  0.02598397,
        0.00262097,  0.00159166], dtype=float32),
 array([-0.01874067,  0.00562666,  0.00983806, ..., -0.00271972,
        -0.02530917, -0.02068929], dtype=float32),
 array([-0.01804586, -0.00070537, -0.0304952 , ..., -0.0283718 ,
        0.00564185, -0.01977405], dtype=float32),
 array([-0.02174005, -0.01807855, -0.02759006, ..., -0.03101528,
        0.01074827, -0.01955673], dtype=float32),
 array([ 0.00279652,  0.01361282, -0.03184221, ..., -0.01509924,
        -0.00120035,  0.00162626], dtype=float32),
 array([-0.022155 ,  0.00387926, -0.02752257, ..., -0.02850369,
        0.0029723 ,  0.0042977 ], dtype=float32),
 array([-0.00415256,  0.0020796 ,  0.01115432, ..., -0.00900922,
        -0.03046259, -0.00698692], dtype=float32)]
```

```
query="I want to watch space movie"
```

```
embeddings_Q= model.encode(query,
                             batch_size=12,
                             max_length=8192, # If you don't need such a long length, you can set a smaller value to speed up the encoding
                             )['dense_vecs']
```

```
embeddings_Q
```

```
array([-0.02240945,  0.02309795, -0.04404929, ..., -0.06079755,
        -0.03096655,  0.051058 ], dtype=float32)
```

```
for i in embedding:
    similarity_score=np.dot(i, embeddings_Q) / (np.linalg.norm(i) * np.linalg.norm(embeddings_Q))
    print(similarity_score)
```

```
0.51576585
0.56447333
0.43669143
0.465387
0.48650634
0.44124305
0.44692978
0.44884613
0.45476976
0.44492936
0.41971424
0.43065858
0.48531336
0.45900872
0.44435847
```

```
#print movie names as well with matching score in descending order
```

```
#howework !!!!
```